IQR

**What is IQR?**

**IQR = Q3 – Q1**

It measures the **spread of the middle 50% of the data**.

- Q1 → 25th percentile
- Q3 → 75th percentile

So, IQR tells us:
**"Where the majority (middle 50%) of students lie."**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **IQR** | 107.0 | 15.1 | 12.1 | 11.0 | 23.5 | 8.31 | 60000.0 |
| **1.5 Rule** | 160.5 | 22.65 | 18.15 | 16.5 | 35.25 | 12.465 | 90000.0 |
| **Lesser** | -106.0 | 37.95 | 42.75 | 44.5 | 24.75 | 45.48 | 150000.0 |
| **Greater** | 322.0 | 98.35 | 91.15 | 88.5 | 118.75 | 78.72 | 390000.0 |

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📈 **IQR Calculation (From Your Data)**

**SSC Percentage (ssc_p)**

- Q1 = 54.5
- Q3 = 75.7

**IQR = 75.7 – 54.5 = 21.2**

**Interpretation:**
The middle 50% of students scored between **54.5% and 75.7%**, indicating **moderate variation** in SSC performance.

**Insight:**

- Middle 50% students scored between **54.5% and 75.7%**
- Moderate spread → **balanced performance**

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**HSC Percentage (hsc_p)**

- Q1 = 60.6
- Q3 = 73.0

**IQR = 12.4**

**Interpretation:**
Scores are more concentrated, showing **less variability** compared to SSC.

**Insight:**

- Scores are more **closely packed**
- Less variation compared to SSC

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**3. Degree Percentage (degree_p)**

- Q1 = 60.9
- Q3 = 72.0

**IQR = 11.1**

**Interpretation:**
Very small spread indicates **consistent academic performance** among students.

**Insight:**

- Very consistent scores
- Indicates **stable academic performance**

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**4. MBA Percentage (mba_p)**

- Q1 = 57.94
- Q3 = 66.25

**IQR = 8.31**

**Interpretation:**
Lowest IQR → **least variation**, meaning most students performed similarly in MBA.

**Insight:**

- Smallest IQR → **least variation**
- Students performed **similarly in MBA**

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**5. Employability Test (etest_p)**

- Q1 = 60.0
- Q3 = 83.5

**IQR = 23.5**

**Interpretation:**
Highest IQR → **large variation in skills**, indicating a **significant performance gap** among students.

**Insight:**

- Largest IQR → **very high variation**

- Big difference in student skills

**Key Finding:**
**Skills vary much more than academic scores**

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**6. Salary**

- Q1 = 240,000
- Q3 = 300,000

**IQR = 60,000**

**Interpretation:**
Most students earn between **2.4L and 3L**, but variation exists due to different job offers.

**Insight:**

- Most students earn between **2.4L and 3L**
- Moderate variation in salary

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📊 **Outlier Detection Using IQR**

Formula:

- Lower Bound = Q1 – 1.5 × IQR
- Upper Bound = Q3 + 1.5 × IQR

**Salary:**

- **Upper Bound = 300,000 + (1.5 × 60,000) = 390,000**
- **Max Salary = 940,000**

**Since 940,000 > 390,000 → Extreme Outlier**

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**Overall Insights**

**Academic Scores**

- IQR is smaller → **consistent performance**

**Employability Test**

- Highest IQR → **large skill gap**

**Salary**

- Moderate IQR but extreme max value
  → **few students earn very high salaries**

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**Final Insights**

- **Academic scores (SSC, HSC, Degree, MBA) show low to moderate variation**
- **Employability test shows high variation → skill gap among students**
- **Salary distribution is affected by outliers, with a few students earning very high salaries**

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**Conclusion**

*The IQR analysis shows that academic performance is relatively consistent among students, while employability skills vary significantly. This variation in skills leads to differences in salary, with a few high-paying outliers affecting the overall distribution.*